SMART PARKING SYSTEM



PRESENTED BY:

REEM MOSTAFA MOHAMED KAMAL
SHAHD EID RAMADAN IBRAHIM
YASMEIN MAHMOUD BAKRY
AZAA MAHMOUD TOHAMY

PRESENTED TO:

DR/ MOKHTAR SAID IBRAHIM



WHY THIS PROJECT?

01

- Real-life application used in malls, hospitals, universities.

02

- Helps organize traffic flow and prevents congestion.

03

- Low-cost but highly functional system.

04

- Electricity cost saving.

PROJECT GOALS :



- Automate car entry and exit using sensors and a servo gate.
- Display available parking slots on an LCD.
- Automatic management for parking areas.
- Displays when the parking is full.
- Trespassing alarm system.
- Optimize electricity by litting pole lights for a specified period during drivers existance and entrance.

HOW THE SYSTEM WORKS? (CONCEPT)

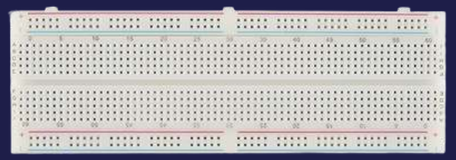
- High-Level Operation.
- IR1 detects car entering.
- IR2 detects car exiting.
- Servo gate opens automatically.
- Slot variable increments or decrements in range of [0,4].
- LCD always shows current available slots.
- Buzzer warns when slot > 4 (trespassing or abnormal condition).
- Light poles turn ON for 4 seconds when car enters or exits.



COMPONENTS LiST:



Arduino Uno



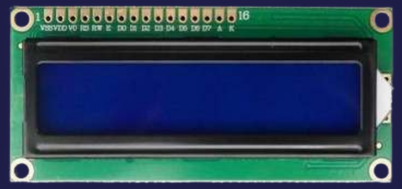
Breadboard



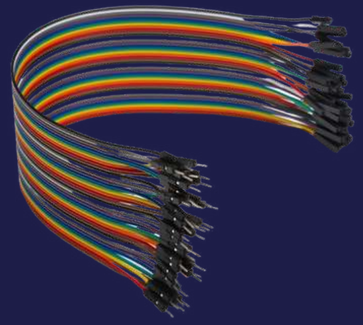
LEDx4



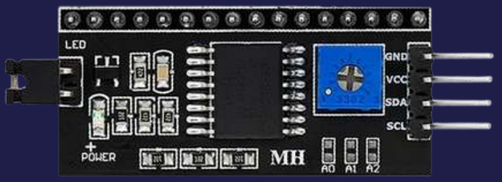
Servo Motor SG90



LCD 16x2



Jumper wires



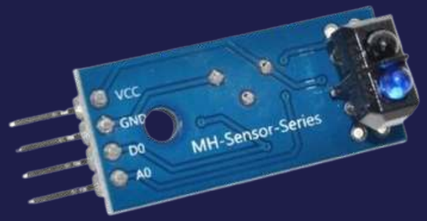
I2C Module



Standard type A
mini type B
USB cable



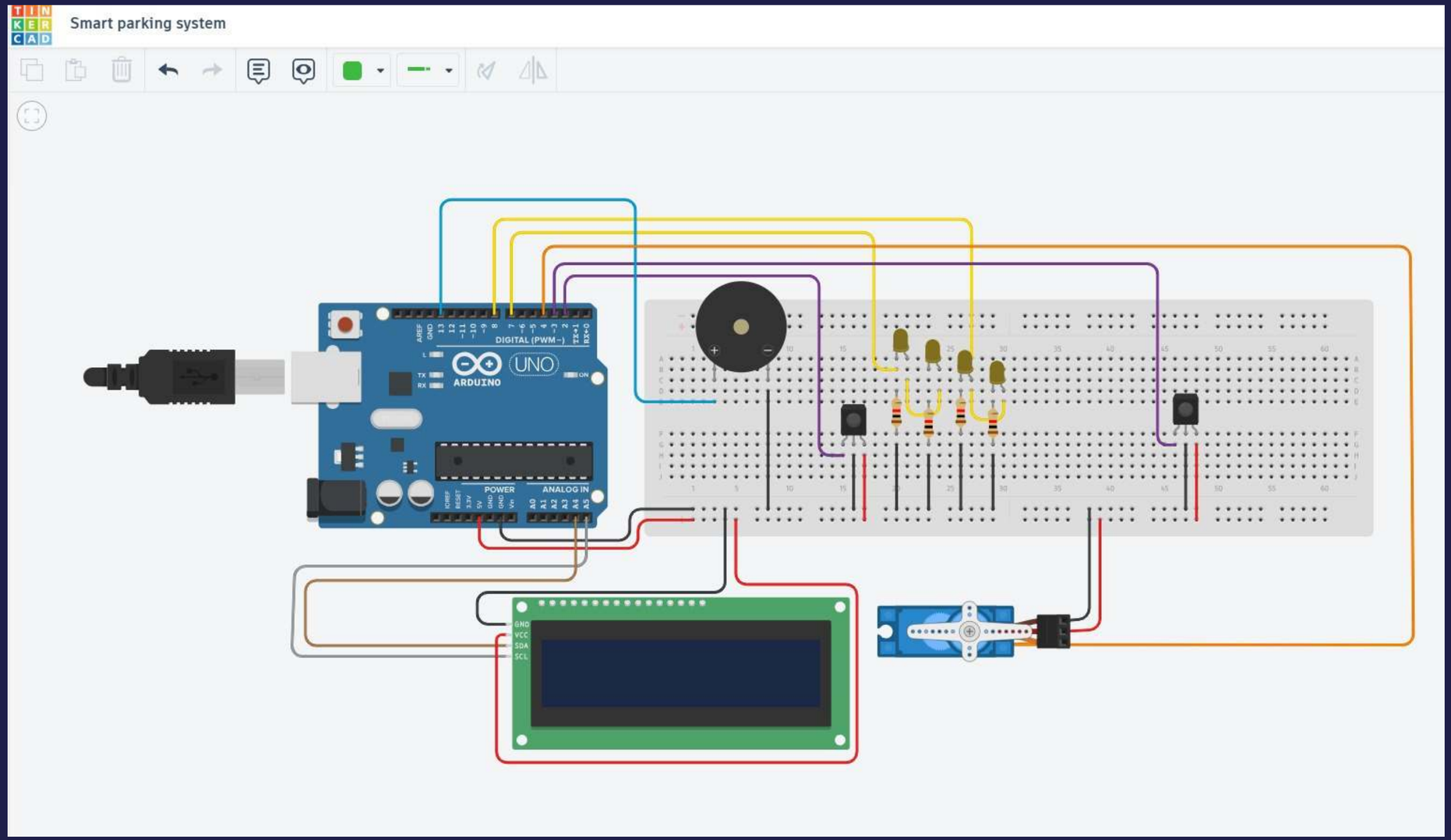
Buzzer



Line Tracker (IR)
Sensors × 2

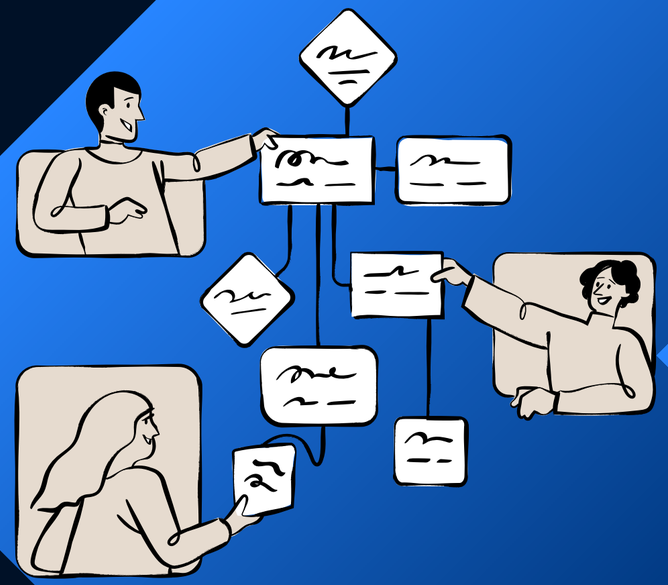
Wiring Diagram and Simulation:

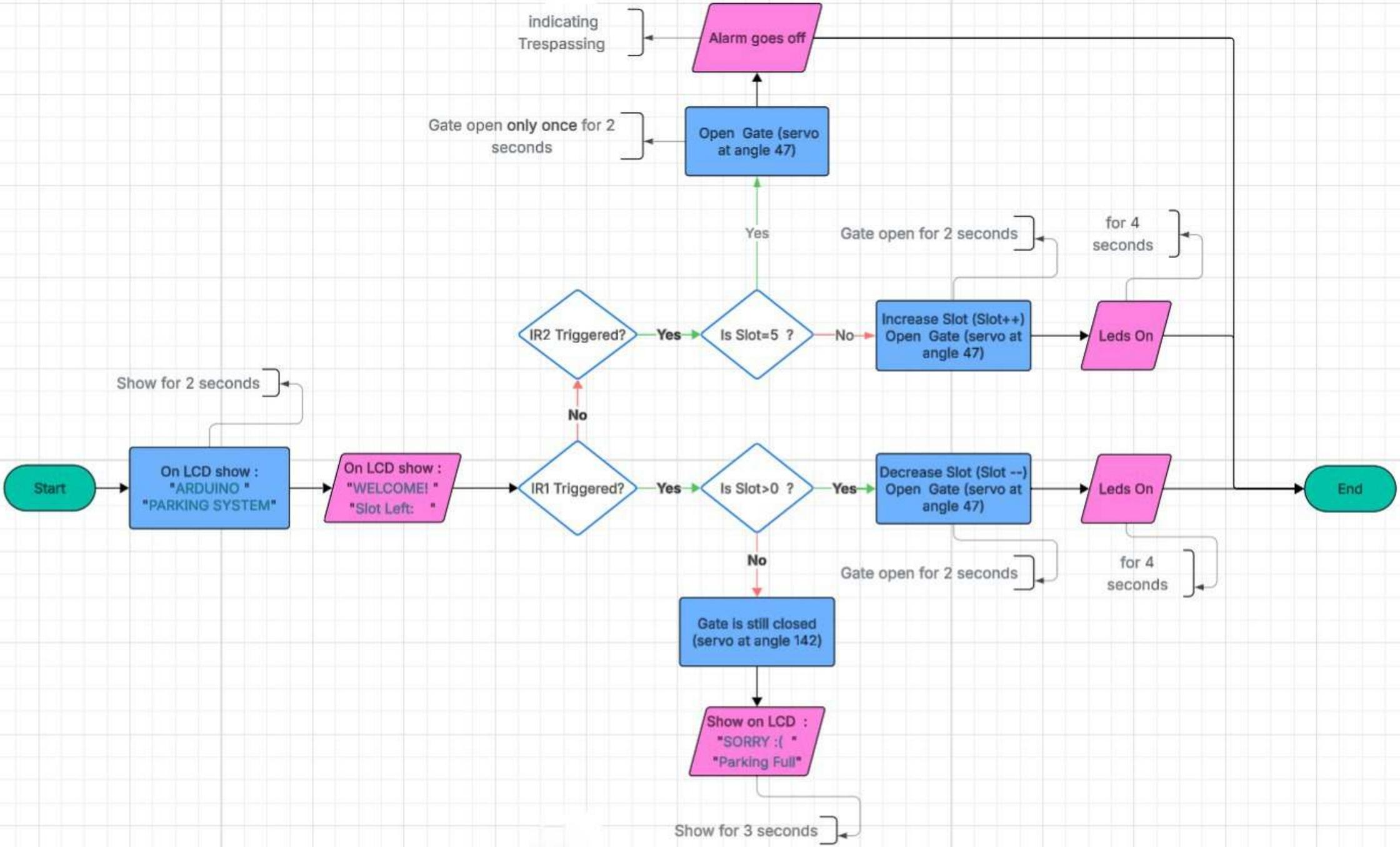
- IR1 → Pin 2
- IR2 → Pin 3
- Servo → Pin 4
- Buzzer → Pin 13
- LEDs → Pins 7, 8
- LCD I2C (SCL) → A5
- LCD I2C (SDA) → A4



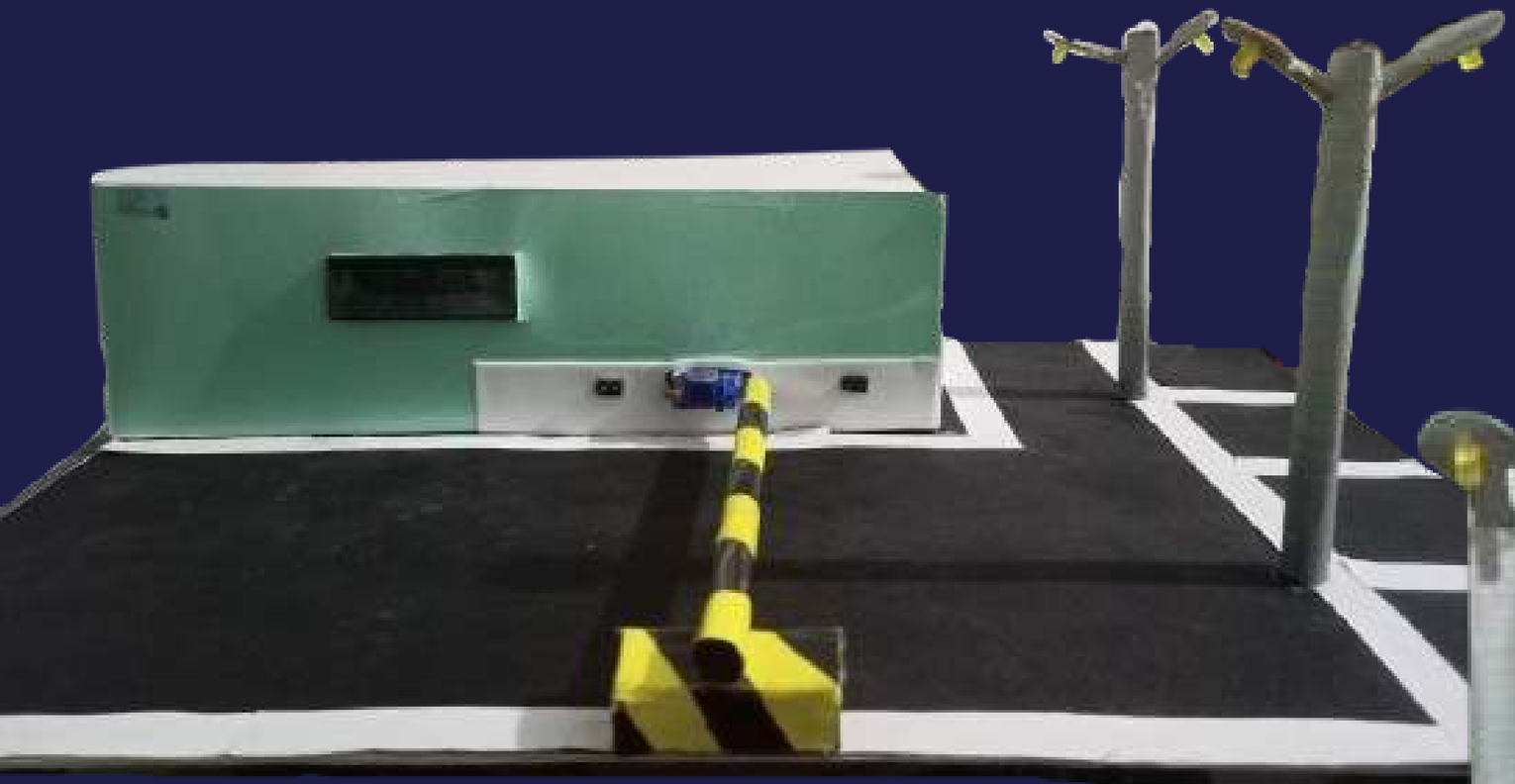
Fritzing diagram using tinkercad

SYSTEM WORKFLOW FLOWCHART




indicating Trespassing
Alarm goes off
Gate open only once for 2 seconds
Open Gate (servo at angle 47)
Yes
Gate open for 2 seconds
for 4 seconds
IR2 Triggered?
Yes
Is Slot=5 ?
No
Increase Slot (Slot++) Open Gate (servo at angle 47)
Leds On
Show for 2 seconds
No
Start
On LCD show : "ARDUINO " "PARKING SYSTEM"
On LCD show : "WELCOME! " "Slot Left: "
IR1 Triggered?
Yes
Is Slot>0 ?
Yes
Decrease Slot (Slot --) Open Gate (servo at angle 47)
Leds On
End
No
Gate open for 2 seconds
for 4 seconds
Gate is still closed (servo at angle 142)
Show on LCD : "SORRY :(" "Parking Full"
Show for 3 seconds

PROJECT PiCTURES







CODE

Arduino Uno

Parking_System.ino

```
//LCD libraries
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2);

//Servo motor libraries
#include <Servo.h>

Servo myservo;


//Defining pins
#define Buzzer  13 //Trespassing alarm

int IR1 = 2;     //IR for enterance gate
int IR2 = 3;     //IR for Exit gate

int ledPin1 = 7; //Light pole1
int ledPin2 = 8; //Light pole2

int Slot = 4;    //Total number of parking Slots

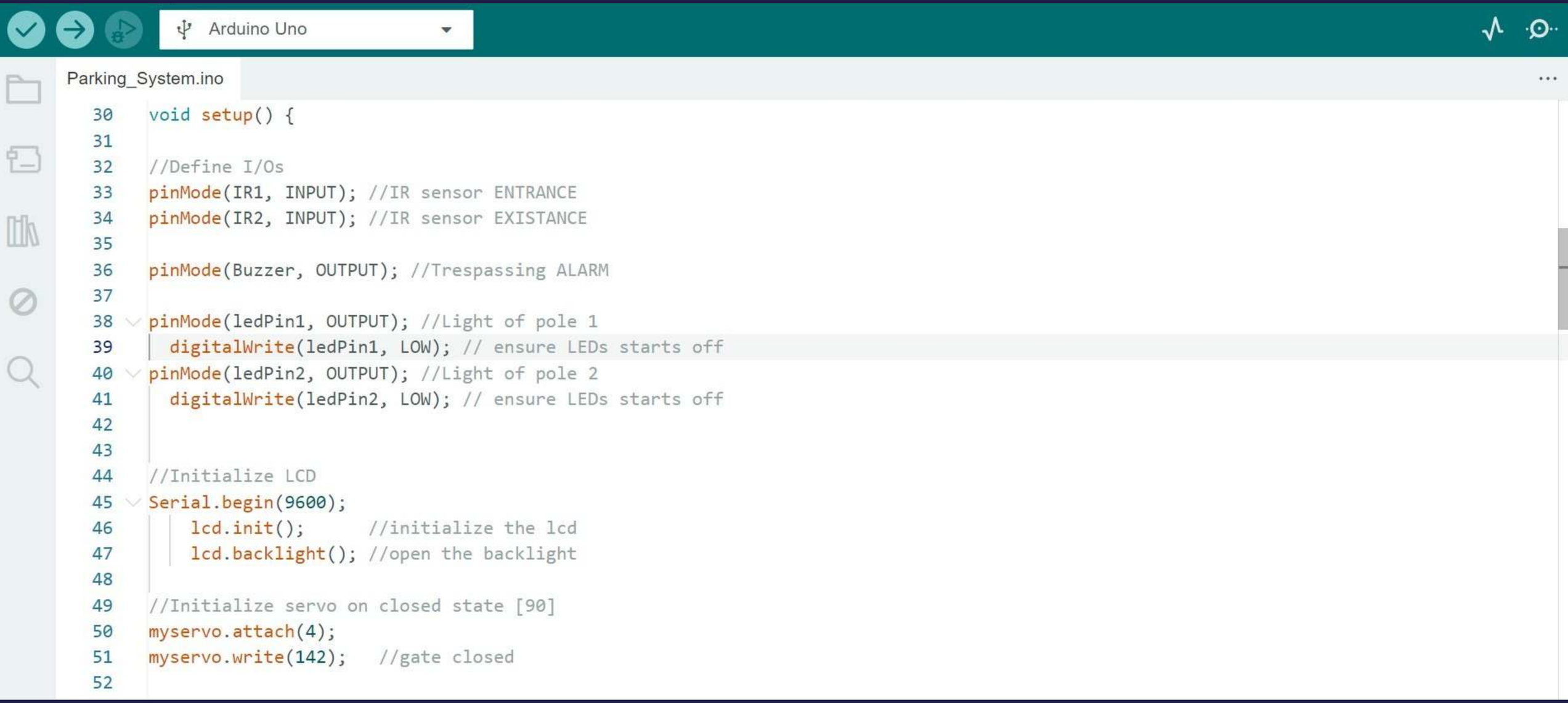
int flag1 = 0;  //flag for enterance gate

int flag2 = 0;  //flag for Exit gate

long ledStart = 0; //var for pole to lit
bool ledOn = false;//flag for pole to lit

void setup() {
```

Output


Arduino Uno
Parking_System.ino
30 void setup() {
31
32 //Define I/Os
33 pinMode(IR1, INPUT); //IR sensor ENTRANCE
34 pinMode(IR2, INPUT); //IR sensor EXISTANCE
35
36 pinMode(Buzzer, OUTPUT); //Trespassing ALARM
37
38 pinMode(ledPin1, OUTPUT); //Light of pole 1
39 digitalWrite(ledPin1, LOW); // ensure LEDs starts off
40 pinMode(ledPin2, OUTPUT); //Light of pole 2
41 digitalWrite(ledPin2, LOW); // ensure LEDs starts off
42
43
44 //Initialize LCD
45 Serial.begin(9600);
46 lcd.init(); //initialize the lcd
47 lcd.backlight(); //open the backlight
48
49 //Initialize servo on closed state [90]
50 myservo.attach(4);
51 myservo.write(142); //gate closed
52

Arduino Uno

Parking_System.ino

```
    lcd.init();      //initialize the lcd
    lcd.backlight(); //open the backlight

//Initialize servo on closed state [90]
myservo.attach(4);
myservo.write(142);   //gate closed

//Start system LCD
lcd.setCursor (0,0);
lcd.print("     ARDUINO    ");

lcd.setCursor (0,1);
lcd.print(" PARKING SYSTEM ");

delay (2000);
lcd.clear();

} //end of setup function


void loop() {
```

Arduino Uno

Parking_System.ino

```
63  } //end of setup function
64
65
66  void loop() {
67
68  //Case_1 [OPEN] : Gate Entrance
69  if(digitalRead (IR1) == 0 && flag1==0){
70              if(Slot>0){
71                          flag1=1;
72                          if(flag2==0){myservo.write(47); Slot--;} //0--->47
73                        }else{
74                        lcd.setCursor (0,0);
75                        lcd.print("    SORRY :(    ");
76                        lcd.setCursor (0,1);
77                        lcd.print("  Parking Full  ");
78                        delay (3000);
79                        lcd.clear();
80                          }
81              //Lit light Poles for ENTRANCE
82              digitalWrite(ledPin1, 1);
83              digitalWrite(ledPin2, 1);
84              ledStart = millis(); //led timer flag takes value of Arduino ON time
85              ledOn = true; //light poles ON [ENTRANCE]
86                                                  }
87
88
```

Arduino Uno

Parking_System.ino

```
86                                                  }
87
88
89  //Case_2 [OPEN] : Gate Exit
90  if(digitalRead (IR2) == 0 && flag2==0){
91                                                  flag2=1;
92                                                  if(flag1==0 && Slot<=4){myservo.write(47); Slot++;} //0--->47
93
94                                                  //Lit light Poles for EXISTANCE
95                                                  digitalWrite(ledPin1, 1);
96                                                  digitalWrite(ledPin2, 1);
97                                                  ledStart = millis();
98                                                  ledOn = true; //light poles ON [EXISTANCE]
99                                                  }
100
101
102 //Case_3 [CLOSE] : Gate Entrance and Gate Exit
103 if(flag1==1 || flag2==1){
104   delay (2000);
105   myservo.write(142); //90--->142
106   flag1=0, flag2=0;
107
108 }
109
110
111 //LCD message for ENTRANCE AVALAIBLE
112 lcd.setCursor (0,0);
```

Arduino Uno

Parking_System.ino

```
111  //LCD message for ENTRANCE AVALAIBLE
112  lcd.setCursor (0,0);
113  lcd.print("    WELCOME!    ");
114  lcd.setCursor (0,1);
115  lcd.print("Slot Left: ");
116  lcd.print(Slot);
117
118
119  //Trespassing ALARM
120  if (Slot>4){
121   tone(Buzzer,1480,200);
122   delay(370);
123   noTone(Buzzer);
124  }
125
126  //Dim light Poles for ENTRANCE and EXISTANCE
127  if (ledOn && (millis() - ledStart >= 4000)) {
128      digitalWrite(ledPin1, 0);
129      digitalWrite(ledPin2, 0);
130      ledOn = false;//light poles OFF [ENTRANCE] and [EXISTANCE]
131  }
132
133
134
135  }   //End of loop function
136
```

FUTURE ENHANCEMENTS



- Ultrasonic sensor instead of IR.
- App monitoring / IoT dashboard.
- Barrier with DC motor + motor driver.
- Automatic number plate recognition.
- Ticket printing.



THANK YOU!

WE APPRECIATE YOUR
TIME AND INTEREST.